

1. Administrative & Study Identification

Full Study Title: Real-world study on dapagliflozin usage in patients with Heart Failure (HF) in Germany **Short Title/Acronym:** EVOLUTION-HF DEalleEF **Protocol Number:** D1699R00050 **NCT Number:** NCT06336330 **Sponsor Name:** AstraZeneca **Investigational Product:** Dapagliflozin **Phase of Development:** Observational / Non-interventional (Phase IV equivalent) **Medical Monitor:** AstraZeneca Clinical Operations Lead (Germany)

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To determine the demographic and clinical characterization of patients and to assess treatment patterns of dapagliflozin, other HF medications, and glucose-lowering medications in a real-world setting. **Secondary Objectives:** To describe Health-Related Quality of Life (HRQoL), medication adherence, and work productivity using patient-reported outcomes (PROs).

2.2 Background & Rationale To address the gap in real-world evidence for dapagliflozin use in heart failure patients across the entire spectrum of ejection fraction (EF). While dapagliflozin (an SGLT2 inhibitor) recently received a label expansion in Germany for the treatment of HF regardless of EF (HFrEF, HFmrEF, and HFpEF), real-world clinical and patient-reported data remain limited. This study aims to complement existing data under the multinational umbrella EVOLUTION-HF study protocol.

3. Study Design & Methodology

3.1 Design Framework Study Type: Observational (Non-interventional, prospective, longitudinal cohort study) **Control Type:** Single-arm **Blinding:** Open-label **Randomization:** N/A

3.2 Planned Enrollment & Duration Target \$\$\$: 1,000 patients (approx. 700 evaluable) with planned distribution: 40% HFpEF, 20% HFmrEF, and 40% HFrEF. **Number of Sites:** Multiple sites across Germany (in cooperation with the German Society of Cardiologists in Private Practice) **Estimated Study Start:** 25-Apr-2024 **Estimated Primary Completion:** 31-Dec-2026

4. Subject Selection (Eligibility Criteria)

- 4.1 Inclusion Criteria**
1. Age ≥ 18 years as of the study index date.
 2. Confirmed diagnosis of symptomatic chronic heart failure (HF) across any ejection fraction spectrum (HFpEF $\geq 50\%$, HFmrEF $41-49\%$, or HFrEF $\leq 40\%$).
 3. Initiated dapagliflozin treatment 14 to 90 days prior to enrolment in accordance with the local product label.
- 4.2 Exclusion Criteria**
1. Previous treatment with any SGLT2 inhibitor (including dapagliflozin) prior to the index date.
 2. Diagnosis of Type 1 diabetes.
 3. Known hypersensitivity to dapagliflozin or its excipients.
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5. Intervention & Treatment Plan

- 5.1 Investigational Regimen Dosage/Frequency:** Dapagliflozin prescribed in accordance with standard clinical practice and the local product label. **Route of Administration:** Oral. **Duration of Treatment:** 12 months of prospective observational follow-up.
- 5.2 Concomitant & Prohibited Therapy Permitted:** Standard of care HF therapies and glucose-lowering medications as prescribed by the physician. **Prohibited:** Participation requires no prior exposure to SGLT2 inhibitors before the current dapagliflozin initiation.
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6. Study Timeline & Visit Schedule

Visit ID	Window	Key Activities/Procedures
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:---	:---	:---
Screening / Index	Day -90 to -14	Dapagliflozin initiation (Index Date), retrospective data collection of medical history and prior HF status.
Baseline	Day 0	Enrolment, Informed Consent, baseline questionnaires (PROs).
Interim Follow-ups	Months 3, 6, and 9	Data extracted from medical records (healthcare utilization, concomitant meds), PRO questionnaires.
End of Study	Month 12	Final clinical assessment, PRO collection, treatment pattern evaluation.

7. Outcome Measures (Endpoints)

- 7.1 Primary Endpoint Definition:** Assessment of treatment patterns of dapagliflozin alongside other HF and glucose-lowering medications over the 12-month follow-up. **Measurement Method:** Extraction from real-world data sources (patient charts, discharge letters, electronic medical records).
- 7.2 Secondary & Exploratory Endpoints**
1. Change in patient-reported HRQoL, medication adherence, and work productivity.
 - 2.

Evaluation of healthcare utilization and patient needs, expectations, and satisfaction with medical care.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Routine pharmacovigilance and standard clinical monitoring through electronic medical records and patient reporting during follow-ups. **Serious Adverse Events (SAEs):** Reported per local regulatory guidelines for non-interventional post-marketing studies. **Data Monitoring Committee (DMC):** Managed by the sponsor (AstraZeneca) and Contract Research Organization (CRO) in alignment with German Medicines Act (§ 67 AMG).

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Enrolled Cohort (Target $N=1,000$); Evaluable Set (Estimated $N=700$) structured by EF stratifications. **Sample Size Justification:** Sized to provide representative descriptive statistics of real-world treatment patterns across the three HF strata (HF_rEF, HF_mrEF, HF_pEF) in Germany. **Handling of Missing Data:** Appropriate imputation methods for observational PROs and descriptive reporting for missing real-world medical record data.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study is conducted in accordance with the Declaration of Helsinki and local non-interventional study guidelines (§ 67 German Medicines Act). **Monitoring Plan:** CRO and sponsor affiliates assure compliance via site contacts and monitoring of informed consent forms (ICFs) and medical records. **Audit Trail:** Secure electronic data capture (EDC) from source files combined with PRO inputs.

11. Study Identifiers & Governance

Primary ID: D1699R00050 **Registry Number:** NCT06336330 **Sponsor/Lead Organization:** AstraZeneca **Study Title:** EVOLUTION-HF DEa11EF **Official Regulatory Title:** Real-world study on dapagliflozin usage in patients with Heart Failure (HF) in Germany

12. Clinical Framework (Heart Failure Specific)

Primary Condition: Heart Failure (HF_pEF, HF_mrEF, HF_rEF) **Diagnosis Threshold:** Symptomatic chronic HF regardless of ejection fraction **Medical Methodology:** SGLT2 inhibitor (SGLT2i) initiation

Optimization Target: Guideline-directed medical therapy (GDMT) including dapagliflozin

13. Study Procedures & Methodology

Management Pathway: Standard real-world clinical pathway for HF in Germany **Education Protocol (HCP):** Investigator training conducted by CRO during site initiation visits **Education Protocol (Patient):** Real-world standard of care; questionnaires on needs/expectations **Quality Audit Mechanism:** ICF and medical record monitoring by sponsor delegates/CRO

14. Observation & Follow-Up Schedule

Total Duration: 12 Follow-up months **Onsite Visit Cadence:** Routine real-world clinical practice visits **Remote Monitoring Frequency:** Data extraction every 3 months **Checklist Review Interval:** Questionnaires (PROs) administered every 3 months

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				